

Lesson 15 Strong Induction

Prove the following by strong induction.

1. Suppose $b_1 = 3$, $b_2 = 9$, and $b_n = 2b_{n-1} + 3b_{n-2}$, for $n \geq 3$, then for every natural number n ,

$$b_n = 3^n$$

2. Suppose $a_1 = 1$, $a_2 = 1$, and $a_n = 2a_{n-1} - a_{n-2}$, for $n \geq 3$, then a_n is a constant sequence of 1.
3. Let F_n be the Fibonacci sequence: $F_1 = F_2 = 1$, and $F_n = F_{n-1} + F_{n-2}$ for $n \geq 3$. Then for every $n \in \mathbb{N}$, $F_n < 2^n$.
4. Every integer greater than 1 has a prime factor.
5. We can form any amount of postage 18 cents or greater with only 4-cent and 7-cent stamps.

The following problems are optional.

6. (Optional) The n th Fibonacci number is,

$$F_n = \frac{\left(\frac{1+\sqrt{5}}{2}\right)^n - \left(\frac{1-\sqrt{5}}{2}\right)^n}{\sqrt{5}}$$

7. (Optional) Strong induction is powerful and can be “abused”. Following is a classic example of a false proposition with a wrong proof. Find the mistake.

Proposition Any set of n apples have the same color.

Proof If $n = 1$, then the apple clearly has the same color as itself. Suppose that the proposition is true for any set of k apples. Then for a set of $k + 1$ apples, by the induction hypothesis, the first k apples have the same color and the last k apples have the same color. So, the first apple has the same color as the k th apple, which have the same color as the $(k + 1)$ th apple. Hence all the $k + 1$ apples have the same color. By induction, the proposition is true for all n .